

7. INTRODUCTION TO TRANSIENT HEAT CONDUCTION AND LUMPED HEAT CAPACITANCE METHOD

(1) Governing equation for transient heat conduction

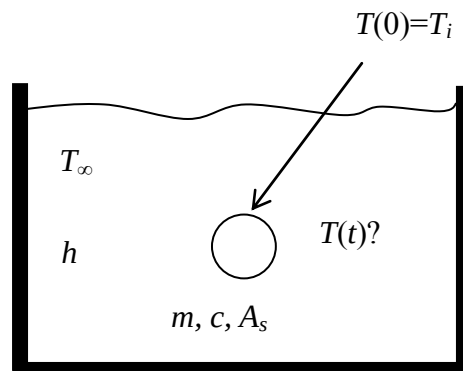
$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q}$$

1-d, constant property: $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$

Note

- Governing equation=PDE → hard to solve
 - Heat capacity becomes important as the heat storage term now survives
 - Thermal diffusivity (important only in transient problem)
- Thermal conductivity vs. Thermal diffusivity?

(2) Lumped heat capacitance method $T(t)$



Motivation: PDEs are hard to solve → consider only temporal variation of temperature, assume that the temperature distribution is uniform in a solid object → Lumped heat capacitance method

$$\rho V c \frac{dT}{dt} = h A_s (T - T_\infty)$$

$$\frac{T - T_\infty}{T_i - T_\infty} = \exp\left(-\frac{h A_s}{\rho V c} t\right) = \exp(-Bi \cdot Fo)$$

Note) $\tau \equiv \frac{\rho V c}{h A_s}$ characteristic time for temperature change $\frac{T - T_\infty}{T_i - T_\infty} = \exp\left(-\frac{t}{\tau}\right)$

<Validity of the lumped heat capacitance analysis>

The above approach seems effective but is it ok to neglect spatial temperature distribution?

- If the temperature difference in the solid is small compared with the temperature difference between the solid object and the ambient
- If the conduction thermal resistance is compared small compared with the convection thermal resistance, the temperature difference in the solid is small compared with the temperature difference between the solid object and the ambient

- $\frac{R_{t,cond}}{R_{t,conv}} = \frac{\frac{L_c}{kA}}{\frac{1}{hA}} = \frac{hL_c}{k} \ll 1$, where $Bi \equiv \frac{hL_c}{k}$

- If Bi number is much smaller than 1, the approach is valid.
- $Bi < 0.1$ → lumped heat capacitance method

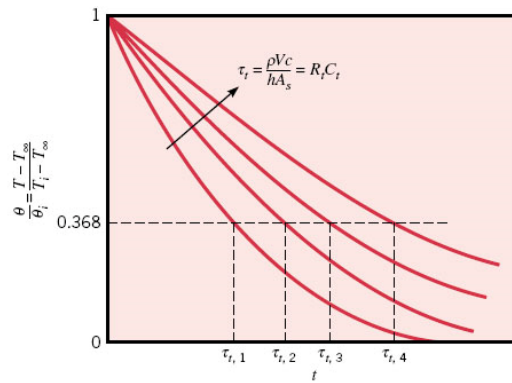


FIGURE 5.2 Transient temperature response of lumped capacitance solids for different thermal time constants τ_t .

(3) Dimensionless form of governing equation

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$T(x,0) = T_i \quad T(0,t) = T_\infty \quad -k \frac{\partial T}{\partial x} \Big|_{x=L} = h[T(L,t) - T_\infty]$$

$$\theta \equiv \frac{T - T_{\infty}}{T_i - T_{\infty}} \quad x^* \equiv \frac{x}{L} \quad t^* \equiv Fo \equiv \frac{\alpha t}{L^2} \text{ (Fourier number)}$$

$$\boxed{\frac{\partial \theta}{\partial (Fo)} = \frac{\partial^2 \theta}{\partial x^{*2}}}$$

$$\theta(x^*, 0) = 1 \quad \theta(0, Fo) = 0 \quad -\left. \frac{\partial \theta}{\partial x^*} \right|_{x^*=1} = Bi \cdot \theta(1, Fo)$$

Problem completely described by $\theta, x^*, Bi \equiv \frac{hL}{k}, Fo \equiv \frac{\alpha t}{L^2}$

● Biot number

$$\blacksquare Bi \equiv \frac{hL_c}{k}$$

$$\blacksquare Bi = \frac{\frac{L_c}{kA}}{\frac{1}{hA}} = (\text{conduction thermal resistance}) / (\text{convection thermal resistance})$$

● Fourier number

$$\blacksquare Fo \equiv \frac{\alpha t}{L_c^2} = (\text{diffusion length})^2 / (\text{characteristic length of the system})^2$$

■ Dimensionless time

$$\blacksquare Fo = \frac{kL_c^2 \frac{\Delta T}{L_c}}{\frac{\rho c_p L_c^3 \Delta T}{t}}$$

= (rate of heat conduction) / (rate of heat storage for a time duration t)

Review problems

(Q1) In the above cooling problem, the lumped body experiences a negligibly small temperature change after sufficient time has elapsed. What is the characteristic time (time constant)?

(Q2) What is the physical meaning of very large/small Fourier number?